

Zero-shot Transfer of Single-Crane DRL Policies for Multi-Crane Container Retrieval

PhD Candidate — AI/ML/DL/Optimization

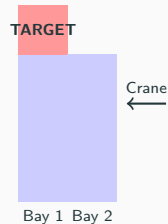
Problem: Container Retrieval Problem

CRP: NP-hard optimization in container terminals

- Yard: B bays $\times R$ rows $\times T$ tiers
- N containers vi retrieval priority $\{1, \dots, N\}$
- Cn relocate blocking containers retrieve
- **Mc tiêu:** Minimize total working time

Gii hn ca SOTA:

- Shin et al. (2026, TR-C): DRL cho **single-crane**
- 7.8% gap vs lower bound trên Lee benchmark



Thực tế

Real-world terminal vận hành **2–3 cranes** trên cùng yard block để đáp ứng throughput.

Không trong

- Toàn bộ CRP research (15+ năm): chỉ single-crane
- M-CRP chưa có nghiên cứu
- Chưa có lower bound, benchmark, method cho multi-crane
- Không biết single-crane policy có zero-shot transfer hay không

Research Question:

Can a single-crane DRL policy be transferred zero-shot to multi-crane environments?

Contributions (5 Claims)

[C1] **Formal M-CRP definition** + NP-hardness proof

[C2] **M-CRP Lower Bound (Theorem 3)** — MCRP

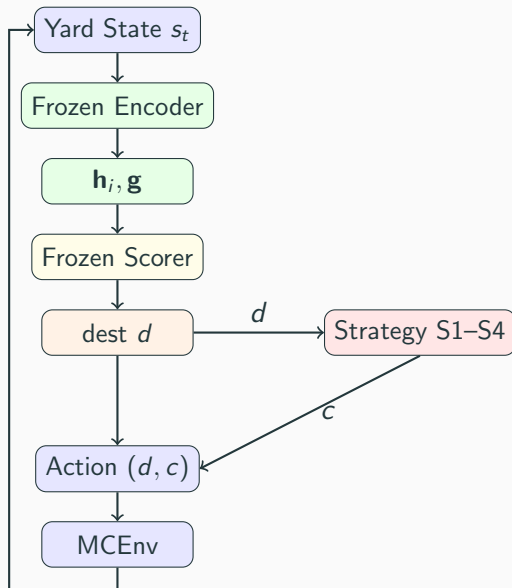
$$\text{MCRP} = \text{retrieval} + \frac{\text{relocation}}{C} + \text{interference}$$

[C3] **4 Crane Assignment Strategies** for zero-shot transfer

[C4] **First empirical evaluation:** 140 instances, 1,120 runs

[C5] **Public benchmark dataset** (CC BY 4.0)

Method: Decoupled Inference Architecture



4 Crane Assignment Strategies

Strategy	Mechanism	Complexity
S1 RoundRobin	Cyclic task assignment	$O(1)$
S2 ZoneSplit	Static bay zones per crane	$O(B)$
S3 LoadBalance	Min-count task assignment	$O(C)$
S4 GreedyOptimal	1-step lookahead cost min	$O(C^2)$

Key design:

- S1, S3: **spatially unaware** (baselines)
- S2, S4: **spatially aware** (leverage crane position)

Experimental Setup

Dataset:

- 140 M-CRP instances t Lee benchmark
- 2-3 cranes, 2 yard scales
- Random + upside-down distributions

Config:

- Model: Shin et al. epoch(100) — public
- 0 GPU (CPU laptop only)
- 1,120 total runs (≈ 60 phút)
- Backward compatibility verified: $< 2\%$ diff

Single-Crane Results

Method	Mean Gap(%)	Min(%)	Max(%)
Random	68.72	47.16	104.26
Kim2016	25.02	13.27	37.52
Lin2015	10.28	5.55	14.14
ZeroShot (ours)	5.63	4.27	7.04

ZeroShot outperforms Lin2015 (SOTA heuristic) by 4.65pp!

Multi-Crane Results

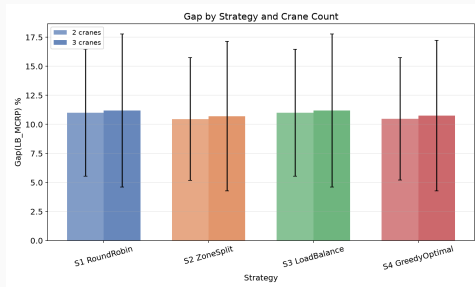
Strategy	C=2 gap(%)	C=3 gap(%)	Intf(C=2)	Intf(C=3)
S1 RoundRobin	10.99	11.19	105.9	147.2
S2 ZoneSplit	10.46	10.71	0.5	0.7
S3 LoadBalance	10.99	11.19	105.9	147.2
S4 GreedyOptimal	10.48	10.75	0.0	0.0

Key findings:

- S2 ZoneSplit: **best gap** + **~ 0 interference** + **$O(B)$ complexity**
- S2 vs S4: không significant difference ($p > 0.1$)
- Spatial awareness $>$ task balancing

Results by Yard Scale

Scale	S1(%)	S2(%)	S4(%)
C=2			
Small (1-4 bays)	9.26	9.04	9.02
Medium (6-10 bays)	13.60	12.58	12.68
C=3			
Small (1-4 bays)	8.43	8.24	8.34
Medium (6-10 bays)	15.33	14.41	14.36



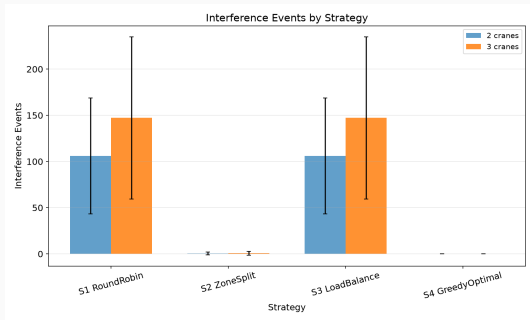
Spatial strategies improve as yard scale grows

Statistical Significance

Pair	C=2 p	C=3 p	Significant?
S2 vs S1	< 0.001	< 0.001	Yes
S4 vs S1	< 0.001	< 0.001	Yes
S2 vs S4	0.128	0.534	No

- **S2, S4 significantly outperform S1, S3** (Wilcoxon, $p < 0.001$)
- S2 vs S4: **no significant difference** $\Rightarrow$ simple ZoneSplit is sufficient

Interference Analysis



Interference reduction:

- S1: 105.9 events ($C=2$)
- S3: 105.9 events
- **S2: 0.5 events (>99.5% reduction)**
- S4: 0.0 events

Spatial awareness eliminates interference without complex coordination!

Why ZoneSplit Works

Key insight

Single-crane DRL policy hc cách relocate containers n stacks gn ó. ZoneSplit align vi behavior này: mi crane operate trong zone riêng.

Practical recommendations:

- **ZoneSplit (S2)** recommend cho deployment
- 0 GPU training cost
- $O(B)$ complexity per decision
- Align vi existing terminal zoning practices

Limitations

1. **Sequential simulation** — serializes crane actions
 - Speedup $C2 \rightarrow C3$: only $1.01\times$ (parallel model: $\approx 1.5\times$)
 - Affects all strategies equally — no bias
2. **LB looseness** at $C > 1$ — negative gaps observed
3. **Single-crane verification** — subset of full benchmark

Future work:

- Fine-tuning policy on multi-crane data
- Multi-agent RL from scratch
- Parallel simulation model

Conclusion

1. **Zero-shot transfer works** — 10.5% gap without retraining
2. **ZoneSplit best** — lowest gap, 0 interference, $O(B)$
3. **Spatial awareness > task balancing**
4. **First M-CRP study** — definition, bound, benchmark

Code + Dataset + Paper:

<https://github.com/sutobode/Master>

Thank You!

Questions?